Douglas County Sample — Shopping Center / Retail

Traffic Impact Study

Prepared August 16, 2026 · 33.7404, -84.7519

SIS sample draft — full tier

Traffic Impact Study

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EXECUTIVE SUMMARY

This report presents the analysis of anticipated traffic impacts associated with the proposed Douglas County Sample — Shopping Center / Retail located within Atlanta MSA, Georgia. The study evaluates 13 intersections within a 1.25-mile study radius using methodology consistent with the Highway Capacity Manual 6th Edition and public-data trip-generation screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716). Trip generation is calculated for land use 820 (Shopping Center / Retail) at a development size of 85 1,000 sqft GLA.

Findings:

• No intersections within the study network are projected to drop one or more LOS during the build conditions.

• No improvements are necessary to maintain the Level of Service standard (LOS D) within the study network.

13	0	0	0.1s
INTERSECTIONS	LOS DROPS	AT LOS E/F	WORST Δ DELAY

1.0 PROJECT DESCRIPTION

1.1 Introduction

This report presents the analysis of the anticipated traffic impacts associated with the proposed Douglas County Sample — Shopping Center / Retail, located within Atlanta MSA, Georgia. Analysis follows methodology consistent with Georgia Department of Transportation (GDOT), Atlanta Regional Commission (ARC), and Georgia Regional Transportation Authority (GRTA) guidance.

1.2 Site Plan Review

Project name	Douglas County Sample — Shopping Center / Retail
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	33.7404°, -84.7519°
Opening year	2027
Region	Atlanta MSA

1.3 Site Access

Site access analysis for proposed driveways is not included in this screening-level analysis. Driveway-level ingress/egress evaluation per GRTA Site Plan Guidelines should be prepared separately based on the final site plan.

1.4 Bicycle and Pedestrian Facilities

Existing bicycle and pedestrian facility inventory within the study area should be confirmed against current GDOT and local agency mapping. ARC-programmed bicycle and pedestrian improvements per the Regional Transportation Plan (RTP) should be reviewed during the methodology meeting.

1.5 Transit Facilities

No transit stops detected within 0.25 mi of the site via the live Transit.land / OSM Overpass query. Transit service area should be confirmed against current MARTA, GRTA Xpress, CCT, GCT, and local transit operator route maps. Proximity to transit influences trip-mode reductions under ARC's Air Quality Benchmark.

2.0 TRAFFIC ANALYSIS METHODOLOGY AND ASSUMPTIONS

2.1 Growth Rate

Background traffic growth is applied at 1.5% per year over 1 year. This rate is consistent with GDOT historical traffic count growth observed along adjacent roadways within the study area. For DRI submittals, the growth rate is typically agreed upon during the pre-application methodology meeting with GRTA, ARC, GDOT, and the local jurisdiction.

2.2 Traffic Data Collection

Intersection capacity analysis uses calibration data from the GDOT 511 NaviGAtor system, including live incident, camera, and signal data feeds. Per-intersection delay calibration is updated hourly from the 7-day rolling incident archive. For formal submittal, supplementary peak-hour turning movement counts conducted within the most recent 12 months are recommended.

2.3 Detailed Intersection Analysis

Level of Service (LOS) is calculated per the Highway Capacity Manual 6th Edition, Chapter 19 (Signalized Intersections), Equation 19-13 (control delay) and Equation 19-50 (95th-percentile queue). LOS is reported for each affected intersection per HCM 6th Ed. Exhibit 19-8 thresholds: A ≤10s · B ≤20s · C ≤35s · D ≤55s · E ≤80s · F >80s of average control delay per vehicle.

3.0 STUDY NETWORK

3.1 Gross Trip Generation

Gross trip generation is calculated per the public-data screening rates (NHTS 2017 / SANDAG 2002 / NCHRP 716) for land use 820 (Shopping Center / Retail) at the proposed development size of 85 1,000 sqft GLA. Average rates are used where fitted-curve equations are not available.

Period	Entering trips	Exiting trips
Daily	3,400	3,400
AM peak hour	166	106
PM peak hour	326	354

Trip Distribution by Time of Day

Estimated within-day distribution of the 6,800 gross daily trips and the resulting on-site accumulation. Within-day distribution from NHTS 2022 shopping / errand / meal trip start times, cross-checked against public-data time-of-day patterns for LU 820 / 850. Activity builds to a midday plateau and a stronger late-afternoon/evening peak; on-site occupancy peaks midday through the evening.

Figure — Inbound / Outbound Trips by Start Time

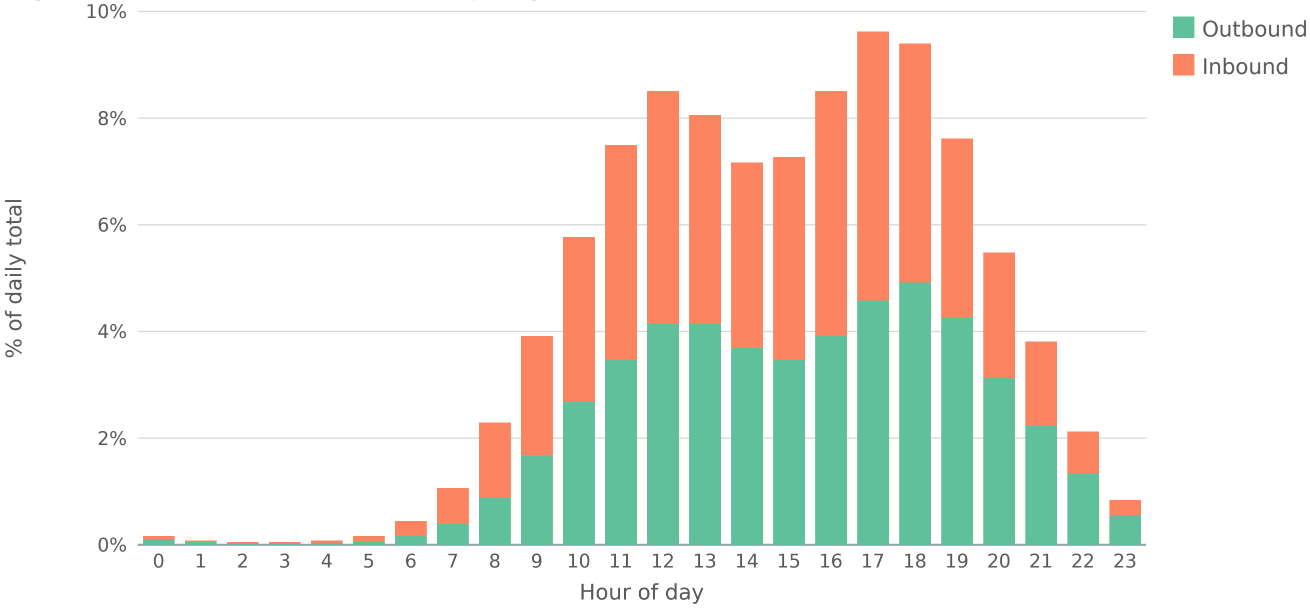
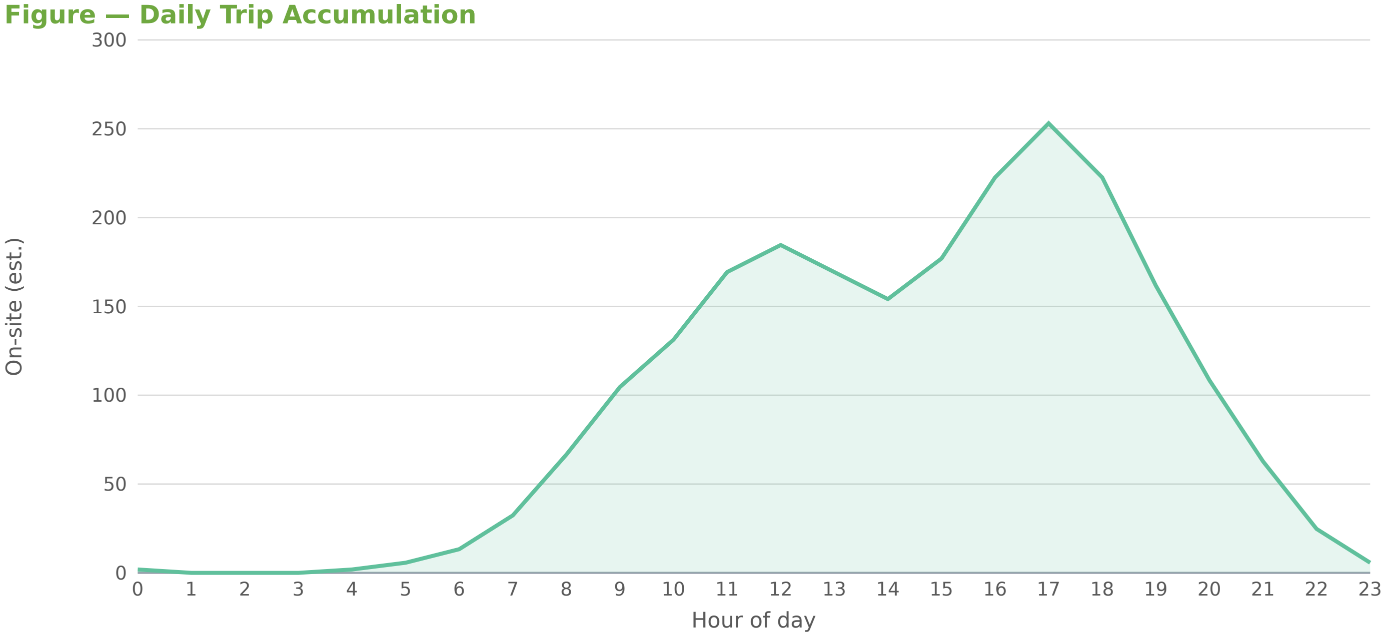


Figure — Daily Trip Accumulation



Peak on-site accumulation ~253 at 17:00, from 6,800 gross daily trips on the retail within-day profile. Screening estimate; not a substitute for a calibrated time-of-day model.

3.2 Trip Distribution

Directional distribution and assignment of new project trips is based on the existing roadway network geometry, proximity to project access points, and engineering judgment. For formal DRI submittal, distribution percentages should be agreed upon during the methodology meeting with GRTA, ARC, GDOT, and the local jurisdiction.

3.3 Level of Service Standards

Per GDOT and GRTA convention, the Level of Service standard for all intersections and roadway segments within the study network is LOS D. Where an intersection or segment currently operates at LOS E or F during the existing peak period, the LOS standard for that period becomes LOS E, consistent with GRTA's Letter of Understanding.

3.4 Study Network Determination

The study network covers all signalized intersections within a 1.25-mile radius of the project site. For

DRI-level submittal, GRTA's 7-percent rule (which extends the network to any intersection or segment where project-generated trips exceed 7 percent of the service volume) should be applied; this screening-level analysis applies the radius-based criterion as a starting point.

3.5 Existing Facilities

Affected intersection	Distance (mi)	Existing LOS	Existing delay (s)
Near Rose Avenue	0.74	B	15.6
Douglas Boulevard & I 20	0.77	B	16.2
Rose Avenue & GA 5	0.86	B	15.2
Near I 20	0.90	B	15.3
Near GA 5	0.91	B	15.8
Signal 20.8mi W of CBD (33.7544, -84.7492)	0.98	B	18.0
GA 5 & I 20	0.99	C	22.2
Near I 20	1.00	B	16.5
Signal 21.3mi W of CBD (33.7259, -84.7572)	1.05	B	15.8
GA 5 & I 20	1.07	B	17.7
Signal 21.3mi W of CBD (33.7256, -84.7576)	1.08	B	16.4
Signal 20.1mi W of CBD (33.7301, -84.7373)	1.10	B	15.5
Near GA 5	1.22	B	16.1

4.0 TRIP GENERATION

Net new trips applied to the study network are calculated by subtracting pass-by capture and internal capture from the gross trip generation. Gross trip rates are drawn from public data (SANDAG 2002 / NHTS 2017 / NCHRP 716); a submittal-grade study should confirm rates against the jurisdiction-approved source.

Independent variable	1,000 sqft GLA (ksf)
Rate basis	Public data — SANDAG 2002 vehicular trip-generation guide
Pass-by capture applied	30%
Internal capture applied	0%
Background growth applied	1.5% per year over 1 year(s)
Weather condition	clear

Period	Raw	Pass-by	Int. cap.	External	In	Out
AM Peak Hour	272	20	0	216	132	84
PM Peak Hour	680	204	0	408	196	212
Saturday Midday	748	56	0	594	297	297
Daily Total	6,800	510	0	5,397	2,698	2,699

The external / net-external column is net-new vehicle trips assigned to the roadway = (gross − pass-by − internal-capture) × auto-mode share (85.7% for this metro). Walk / bike / transit person-trips are excluded from off-site assignment, so In + Out equals the external column.

5.0 TRIP DISTRIBUTION AND ASSIGNMENT

Net new trips are assigned to the study network proportionally to signal proximity and approach geometry. The per-intersection trip allocation for each affected signal is reflected in the Section 6.0 Traffic Analysis tables below.

5.1 Trip Distribution — Gravity Model

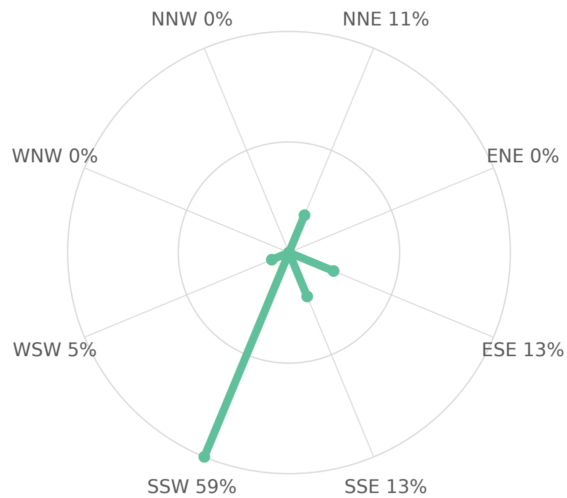
The trip distribution uses the gravity model (term = M / (d^1 · d_site); mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector	Share of project trips
Northeast (NNE + ENE)	11%
Southeast (ESE + SSE)	25%
Southwest (SSW + WSW)	64%
Northwest (WNW + NNW)	0%

Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
GA 5 & I 20	SSW	0.99	1,101	0.2	16.70%
Signal 20.8mi W of CBD (33.7544, -84.7492)	NNE	0.98	707	0.1	10.76%
GA 5 & I 20	SSW	1.07	669	0.1	9.90%
Douglas Boulevard & I 20	SSE	0.77	470	0.1	7.65%
Near I 20	ESE	1.00	503	0.1	7.61%
Signal 21.3mi W of CBD (33.7256, -84.7576)	SSW	1.08	499	0.1	7.37%
Near GA 5	SSW	1.22	459	0.1	6.48%
Near GA 5	SSW	0.91	400	0.1	6.23%
Near Rose Avenue	SSW	0.74	373	0.1	6.14%
Signal 21.3mi W of CBD (33.7259, -84.7572)	SSW	1.05	403	0.1	6.01%
Signal 20.1mi W of CBD (33.7301, -84.7373)	ESE	1.10	356	0.1	5.22%
Near I 20	SSE	0.90	320	0.0	4.99%
Rose Avenue & GA 5	WSW	0.86	312	0.0	4.94%

13 study-area zone(s). Screening-grade distribution; not a calibrated regional model.

Figure — Directional Distribution of Project Trips



Screening-grade directional distribution of net new project trips by compass octant (spoke length ∝ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone

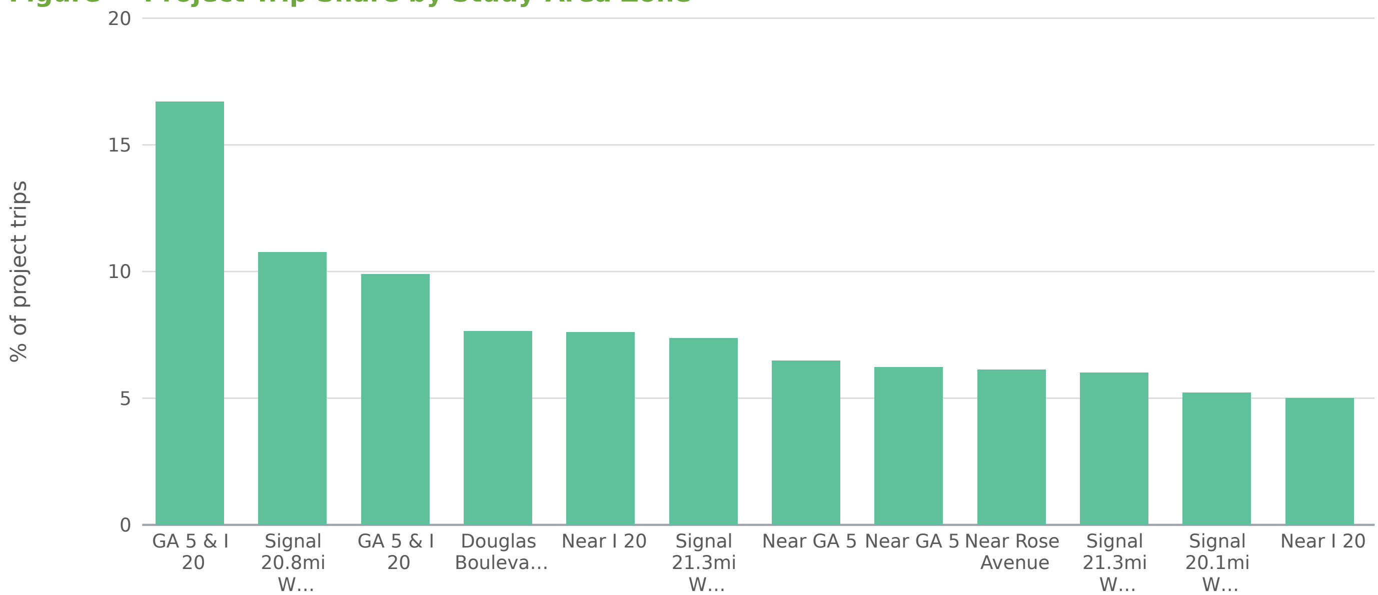
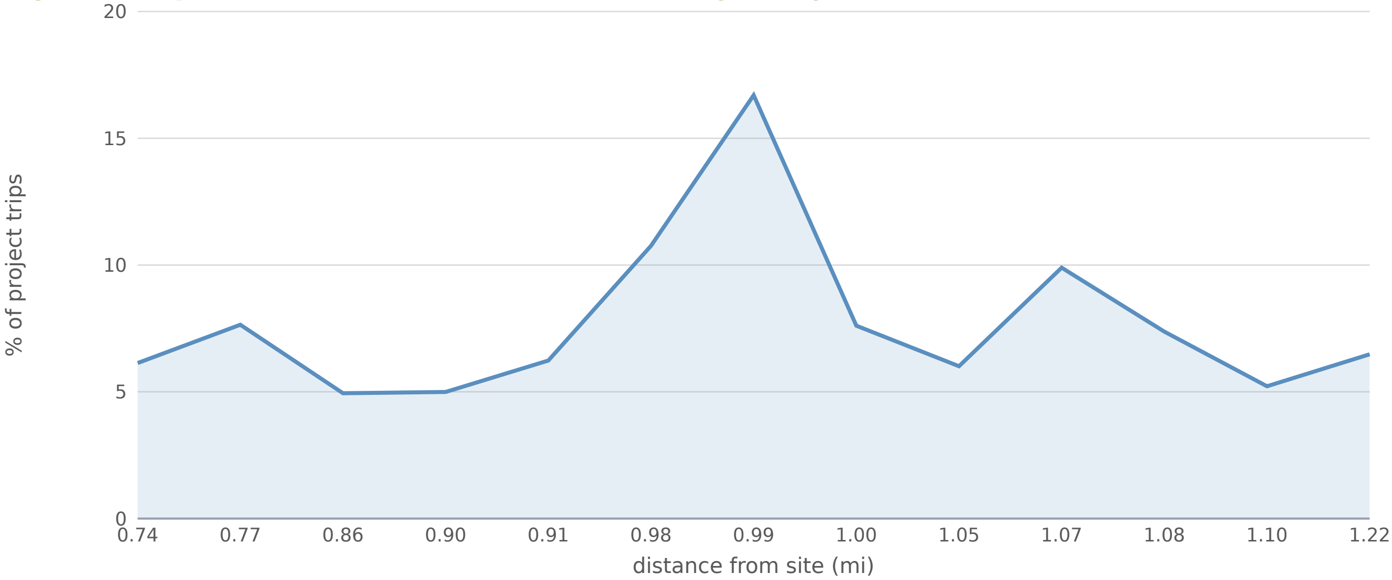


Figure — Trip Share vs. Distance from Site (Gravity Decay)



5.2 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
Near Rose Avenue	SSW	0.74	5	10	19.6%
Douglas Boulevard & I 20	SSE	0.77	5	9	17.6%
Rose Avenue & GA 5	WSW	0.86	3	6	11.8%
Near I 20	SSE	0.90	2	4	7.8%
Near GA 5	SSW	0.91	2	4	7.8%
Signal 20.8mi W of CBD (33.7544, -84.7492)	NNE	0.98	2	3	5.9%
GA 5 & I 20	SSW	0.99	2	3	5.9%
Near I 20	ESE	1.00	1	3	5.9%
Signal 21.3mi W of CBD (33.7259, -84.7572)	SSW	1.05	1	2	3.9%
GA 5 & I 20	SSW	1.07	1	2	3.9%
Signal 21.3mi W of CBD (33.7256, -84.7576)	SSW	1.08	1	2	3.9%
Signal 20.1mi W of CBD (33.7301, -84.7373)	ESE	1.10	1	2	3.9%
Near GA 5	SSW	1.22	0	1	2.0%

Project trips assigned to each study intersection from the distribution shares above: the directional shares orient the loading toward the high-share sectors, distance-decayed to each approach.

6.0 TRAFFIC ANALYSIS

Four scenarios are evaluated at each affected intersection per GRTA / GDOT convention for projects exceeding screening thresholds: (1) Existing — current-year volumes from the GDOT 511 system, no growth applied; (2) No-Build at opening year 2027 — existing volumes grown at 1.5%/yr over 1 year(s); (3) Build at opening year 2027 — No-Build volumes plus project external trips at the assigned distribution; (4) 20-Year Design Year (2047) No-Build and Build — opening-year volumes compounded another 20 years at the same growth rate, project trips at full build-out unchanged.

Intersection	Existing	Opening NB	Opening Bld	Design NB	Design Bld	Δ delay (s)
Near Rose Avenue	B	B	B	B	B	0.1
Douglas Boulevard & I 20	B	B	B	B	B	0.0
Rose Avenue & GA 5	B	B	B	B	B	0.0
Near I 20	B	B	B	B	B	0.0
Near GA 5	B	B	B	B	B	0.0
Signal 20.8mi W of CBD (33.7544, -84.7492)	B	B	B	C	C	0.0
GA 5 & I 20	C	C	C	C	C	0.0
Near I 20	B	B	B	B	B	0.1
Signal 21.3mi W of CBD (33.7259, -84.7572)	B	B	B	B	B	0.0
GA 5 & I 20	B	B	B	B	B	0.1
Signal 21.3mi W of CBD (33.7256, -84.7576)	B	B	B	B	B	0.0
Signal 20.1mi W of CBD (33.7301, -84.7373)	B	B	B	B	B	0.0
Near GA 5	B	B	B	B	B	0.0

No improvements are necessary to maintain the Level of Service standard (LOS D) within the study network under build conditions.

7.0 IDENTIFICATION OF PROGRAMMED PROJECTS

Review of programmed transportation projects within the study area should consult: GDOT Transportation Improvement Program (TIP), Statewide Transportation Improvement Program (STIP), Atlanta Regional Commission Regional Transportation Plan (RTP), and GDOT's Construction Work Program. This screening analysis does not automatically integrate programmed-projects data; manual review is recommended for any submittal.

8.0 INGRESS/EGRESS ANALYSIS

Per-driveway operational analysis is not included in this screening-level study. Proposed site access driveways should be analyzed individually under build conditions to determine ingress and egress operations, including full-movement vs. left-in/left-out configurations and signal warrant evaluation where appropriate.

9.0 INTERNAL CIRCULATION ANALYSIS

Internal site circulation, parking access, and service-vehicle pathways are dependent on the final site plan and are not included in this screening-level analysis. Internal circulation review should follow the local jurisdiction's site plan review process.

10.0 COMPLIANCE WITH COMPREHENSIVE PLAN ANALYSIS

Compliance with the local jurisdiction's Future Land Use Plan and Comprehensive Plan should be confirmed against the most recent adopted plan and any applicable Neighborhood Planning Unit (NPU) or Special Public Interest (SPI) overlay district designations.

11.0 NON-EXPEDITED REVIEW CRITERIA

Per GA DCA Chapter 110-12-3, DRI submittals must address the eight non-expedited review criteria below. Items marked as auto-computed reflect the engine's deterministic outputs from this analysis; items flagged for verification require coordination with GRTA, ARC, MARTA, GDOT, and the local jurisdiction during the pre-application methodology meeting.

11.1 Quality, Character, Convenience, and Flexibility of Transportation Options

Evaluation of transportation options serving the proposed development site requires inventory of: (a) existing and ARC RTP-programmed bicycle and pedestrian facilities within the AOI; (b) MARTA bus and rail service frequency, span, and stop locations within 1/2 mile of the site; (c) GRTA Xpress and regional commuter-coach service to/from the site; and (d) first/last-mile connections between the site and the nearest fixed-route transit. This inventory should be confirmed against current GDOT, ARC, MARTA, and local-agency GIS layers — required for DRI submittal, not auto-generated.

11.2 Vehicle Miles Traveled

Daily VMT is estimated as the product of net external auto-mode trips and an assumed average trip length of 5 miles for land use 820 (Shopping Center / Retail). This trip-length assumption is conservative and reflects Atlanta-region NHTS/ARC regional-travel-survey literature for the land-use category; the formal DRI submittal should substitute a project-specific value from the ARC Activity-Based Model where available.

VMT reduction component	Value	Daily trips	Daily VMT (mi)
Gross trip generation	—	6,800	34,000
Pass-by capture	30%	−510	−2,550
Internal capture (mixed-use)	0%	−0	−0
Alternative-mode share (non-auto)	22%	−1,384	−6,919
Net new auto trips and VMT	—	4,906	24,531

Reduction sources: pass-by per standard screening methodology; internal capture per ULI Mixed-Use Internal Capture defaults; alternative-mode share from ACS B08301 for Atlanta MSA (78% auto). Assumed average trip length is a screening input — not a calibrated AOI-specific value.

11.3 Relationship Between Location of Proposed DRI and Regional Mobility

The proposed development is located within Atlanta MSA. Connections to the regional mobility network — including the Interstate system, GDOT principal arterials, MARTA heavy-rail corridors, and GRTA Xpress park-and-ride facilities — should be enumerated in the DRI submittal based on direct distance and travel time from the site. This screening analysis does not auto-detect specific interstate corridor proximity; that determination requires manual GIS review against the GDOT functional classification layer and is required for DRI submittal.

11.4 Relationship Between Proposed DRI and Existing or Planned Transit Facilities

Required for DRI submittal: inventory of MARTA heavy-rail stations, MARTA bus routes (with peak headways), GRTA Xpress routes, and any ARC RTP-programmed transit expansion projects with right-of-way intersecting the AOI. Walk-shed analysis (1/4 mile and 1/2 mile) to fixed-route transit stops should be presented as a map exhibit. Not auto-generated — requires MARTA GTFS overlay and pre-application coordination with MARTA Planning.

11.5 Transportation Management Area Designation

Required for DRI submittal: identification of any Transportation Management Association (TMA) with service area covering the project site (e.g., Midtown Transportation, Buckhead REdeux, Perimeter Connects, Cumberland Community Improvement District). TMA membership and trip-reduction program participation should be documented, including any TMA-administered Guaranteed Ride Home, vanpool, or transit-pass-subsidy programs the development will participate in. Not auto-generated — requires lookup against the current ARC TMA service-area map and applicant-side membership confirmation.

11.6 Offsite Trip Reduction and Trip Reduction Techniques

Offsite trip reduction credits applied in this analysis are summarized below. Additional Trip Reduction Program (TRP) measures — including transit subsidies, vanpool/carpool incentives, telework programs, parking cash-out, and bicycle facilities — should be enumerated in the DRI submittal as commitments that further reduce vehicle trip generation beyond the screening-level reductions shown here.

Pass-by capture (PM peak)	30%
Internal capture (mixed-use)	0%
Alternative-mode share (non-auto)	22% of external trips arrive by transit, walking, or cycling
Applicant TRP commitments	To be enumerated in DRI submittal (transit subsidy, vanpool, telework, parking cash-out, bike infrastructure)

11.7 Balance of Land Uses — Jobs/Housing Balance

The jobs/housing balance analysis within the AOI is presented in §12 Area of Influence. ARC's review criterion typically targets a jobs-to-housing ratio between 1.3 and 1.7 for activity centers; values outside that range suggest the AOI is either employment-heavy (commuter-trip generating) or housing-heavy (out-commute generating). Refer to §12 for the AOI tabulation and required Census ACS overlay.

11.8 Relationship Between Proposed DRI and Existing Development and Infrastructure

Infrastructure adequacy is project-specific and depends on the site's utility connections (water/sewer/stormwater capacity), surrounding development pattern, and the local jurisdiction's capital improvement program. The DRI submittal should document: water/sewer service availability and capacity letters from the serving utility; stormwater management approach consistent with the GA Stormwater Management Manual; and consistency with the local jurisdiction's adopted Service Delivery Strategy. Not auto-generated — requires applicant-side utility coordination.

12.0 AREA OF INFLUENCE

The Area of Influence (AOI) for this DRI is defined as the area within a 6-mile radius of the project site, consistent with GRTA's standard AOI definition for DRI review. The AOI is centered on the proposed development located in Atlanta MSA and includes all Census block groups whose centroids fall within the 6-mile buffer.

Required AOI demographic overlay

Per GRTA DRI submittal guidance, the AOI characterization requires the following American Community Survey 5-Year Estimate tables aggregated to block-group geography and clipped to the 6-mile buffer. This screening analysis does not auto-generate the Census overlay — the tables below identify the data sources the DRI consultant must compile for the formal submittal.

ACS Table	Subject	AOI tabulation required
B01003	Total population	Population within 6-mi AOI; growth 2010–2020
B25001	Housing units	Total dwelling units within AOI; vacancy rate
B25075	Owner-occupied home value	Median value; distribution by price bin
B25064	Median gross rent	Median rent; rent-to-income ratio
B19013	Median household income	Median household income within AOI

ACS Table	Subject	AOI tabulation required
B23025	Employment status	Labor force; employed civilian population
C24050	Industry of employed pop.	Employment by NAICS sector — jobs side of jobs/housing
B08301	Means of transportation to work	Drive-alone, carpool, transit, walk, bike, work-from-home mode shares
B08303	Travel time to work	Mean commute time; distribution
LEHD LODES WAC	Workplace area characteristics	Jobs by NAICS sector at block-group resolution (jobs side)
LEHD LODES RAC	Residence area characteristics	Resident workers by industry (housing side)

Required AOI analysis exhibits

The DRI submittal must include the following analysis derived from the data sources above. None are auto-computed at this stage.

Population by Census tract	Map exhibit + tabulation — required
Housing units by tract + tenure	Owner/renter split; required
Employment by NAICS sector (LEHD WAC)	Required — jobs side of balance
Jobs/housing balance ratio	Required — ARC target 1.3–1.7 for activity centers
Median household income vs. median home value/rent	Salary-to-housing affordability comparison — required
Mode share to work (ACS B08301)	Required — compares AOI to regional average
Existing land use within AOI	Map exhibit — required, sourced from local jurisdiction

Note: this screening tool does not fabricate AOI demographics. The DRI consultant must compile the above from the named sources prior to submittal. Reference: O.C.G.A. § 50-8-7.1 review criteria + GRTA DRI Review Procedures.

13.0 ARC AIR QUALITY BENCHMARK

The Atlanta Regional Commission Air Quality Benchmark evaluates a DRI's VMT-reduction performance against a scoring rubric of land-use, transit, and trip-reduction credits. Eligibility for each rubric item is determined below; items marked 'Requires verification' depend on AOI-specific GIS data (transit-station proximity, sidewalk network, TMA service area) that this screening tool does not auto-detect.

Rubric item	Status	Notes
Mixed-use development bonus	Not eligible	land use does not indicate mixed-use programming
Internal capture credit (mixed-use)	Not claimed	No internal-capture credit applied
Pass-by trip credit	Eligible — auto-computed	30% credit applied per standard pass-by screening methodology
Alternative-mode share (transit/walk/bike)	Eligible — auto-computed	22% non-auto per ACS B08301 (Atlanta MSA)
Transit-station proximity (≤ 1/2 mi to MARTA rail)	Requires verification	Distance to nearest MARTA heavy-rail station — manual GIS check required

Rubric item	Status	Notes
Bus-stop proximity (≤ 1/4 mi to MARTA bus)	Requires verification	Distance to nearest MARTA bus stop — manual GIS check required
Continuous pedestrian network	Requires verification	Sidewalk and crossing inventory within 1/2 mi of site — local agency data
Bicycle infrastructure (lane/path/shared-use)	Requires verification	Bike-facility inventory within AOI — ARC RTP + local agency data
TMA membership / TRP commitment	Requires verification	TMA service-area lookup + applicant commitment letter required
Park-and-ride / GRTA Xpress access	Requires verification	Distance to nearest park-and-ride lot — GRTA facility map
Auto-computed VMT reduction (from §11.2)	52%	Sum of pass-by + internal capture + non-auto mode share

Notes: The auto-computed VMT reduction reflects only credits supported by engine data. Verification-required rubric items would add to this figure once confirmed during the methodology meeting with GRTA, ARC, MARTA, and the local jurisdiction. The final ARC Air Quality Benchmark score for DRI submittal is determined by ARC review staff and is not produced by this screening tool.

FINDINGS

• Project will generate 6,800 new daily vehicle trips, with 680 during the PM peak hour (326 inbound / 354 outbound).

• Pass-by credit 30% and internal-capture credit 0% applied at the PM peak (25% of that credit at the AM and Saturday-midday periods, per the industry rule of thumb that off-peak shopping diverts less) before off-site assignment (standard pass-by methodology; ULI Internal Capture).

• Auto-mode share of 86% applied to external trips for Atlanta MSA; 14% of generated trips are assumed to arrive by transit, walking, or cycling and do not load the off-site roadway network. Source: ACS 5-Year Estimates Table B08301 (US) or national-stats equivalent.

• Existing volumes were grown by 1.50%/yr over 1 year (×1.01) to the opening-year horizon.

• 13 signalized intersections fall within the study area; 0 are projected to drop at least one LOS grade after build-out.

• All studied intersections are projected to remain at LOS D or better with build traffic; no formal mitigation is required.

• Conclusions are reported at the applied screening assumptions. Discrete-scenario sensitivity at the formal TIA scoping meeting should test: trip-generation method (rate vs. fitted-curve equation); internal capture and pass-by credit variants; and a ±0.5%/yr background growth band around the applied value.

METHODOLOGY NOTES

• Trip generation uses public-data average rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) for the selected land-use code, computed for AM peak, PM peak, Saturday midday, and daily totals. Saturday-midday rates are estimated as a published industry multiple of the PM peak rate by land-use category.

• Pass-by and internal-capture credits are applied in full at the PM peak (and at 25% of that credit fraction for the AM and Saturday-midday periods) per standard pass-by screening methodology and ULI Mixed-Use Internal Capture defaults; only the residual external vehicle trips are assigned to off-site intersections.

• Existing intersection volumes are grown to the opening-year horizon at the user-supplied annual growth rate (default 1.5%/yr) before the capacity analysis.

• Weather adjustment follows HCM 6th-Edition Ch. 11 (rain/snow capacity reduction): clear 1.00, light rain 0.95, heavy rain 0.86, light snow 0.86, heavy snow 0.70. The factor multiplies the saturation flow at every intersection.

• Off-site impact is screened for all signalized intersections within the study radius (default 0.5 mi) using the four-step travel demand model (FHWA; NCHRP Report 716). Step 1 Trip Generation: public-data average rates (SANDAG 2002 / NHTS 2017 / NCHRP 716) give the site's external (post pass-by / internal-capture) productions. Step 2 Trip Distribution: a production-constrained gravity model $T_j = P \cdot (A_j \cdot F_j) / \Sigma(A_x \cdot F_x)$ allocates trips to surrounding signals, where attractiveness A_j is the signal's through-volume and the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work coefficients a=28507, b=-0.02, c=-0.123) on the travel time t to each signal. Step 3 Mode Choice: a binary logit $P(auto)=1/(1+e^{-(ASC-\lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (a density proxy from surrounding through-volumes) so denser, more transit-served sites split further from auto; only the resulting vehicle trips load the network. Step 4 Route Assignment: a capacity-constrained assignment using the BPR volume-delay function $t = t0 \cdot [1 + 0.15 \cdot (v/c)^4]$ iteratively shifts trips away from over-capacity signals toward less-congested alternatives. Signals lacking AADT data fall back to a constant 5,000 vpd attraction.

• After assignment, all signalized intersections within the study radius are reported in the affected-intersections table. Project-added trips, v/c ratio change, control delay change, LOS change, and 95th-percentile queue are reported for each intersection so the reviewer can assess relative impact. Intersections beyond the study radius are excluded from analysis.

• Auto-mode share is applied per metro before assignment. Suburban-US metros default to 90% auto (ACS 5-Year B08301 median); transit-heavy metros use measured auto-mode share (e.g., NYC 32%, Tokyo 30%, London 38%, San Francisco 47%). Non-auto trips (transit, walking, cycling) do not load the off-site roadway. This is a screening-level adjustment; a real TIS submittal in a transit-heavy market should refine with project-specific TAZ data.

• Candidate signals are de-duplicated within a 45m clustering threshold to prevent OSM divided-arterial splits and way-record artifacts from double-counting a single physical intersection.

• Intersection-level control delay uses the HCM signalized-intersection model d = d1 + d2 (Webster uniform delay + Akçelik/HCM incremental-delay term) with a 90s cycle, g/C = 0.45, 1,800 vphpl saturation flow (× weather factor), 15-minute peak analysis period (T = 0.25 hr) and pretimed-signal incremental-delay factor k = 0.5. Reported control delay is capped at 120 s (LOS F) for screening reliability — the incremental-delay term is not calibrated far above capacity, so oversaturated approaches are reported as LOS F rather than an implausible delay figure; a calibrated design-level analysis (HCS / Synchro) supersedes this screen.

• Approach-level analysis splits each signal's inflow across NB/SB/EB/WB approaches (deterministic per-signal allocation perturbed ±15% from a 30/25/25/20 base) and assigns added trips to each approach by cosine-similarity to the bearing of the project relative to the signal. Per-approach v/c, control delay, LOS, and 95th-percentile back-of-queue length (HCM Eq. 19-50, Q95 ≈ Q1 × 1.65 × 25 ft/veh) are reported.

• Level of Service is assigned from HCM 6th-Edition signalized-intersection control-delay thresholds (Exhibit 19-8): A ≤10s, B ≤20s, C ≤35s, D ≤55s, E ≤80s, F >80s.

• Sensitivity is reported in narrative form per standard TIA practice: trip-generation method (rate vs. fitted-curve equation), discrete internal capture and pass-by credit variants, and a ±0.5%/yr growth-rate band around the applied value. The engine retains an internal stochastic-sensitivity routine (Box-Muller-perturbed trip rate and existing volume) for demo-mode diagnostics; it is not surfaced in the deliverable because TIA sensitivity is conducted through discrete scoping-agreed scenario variants, not statistical perturbation of unmeasured distributions.

• Mitigations are screening-level recommendations sized to the projected delay change, not full Synchro/SimTraffic optimization runs. A formal TIS submittal should validate these recommendations with detailed traffic counts and signal-timing analysis.

• Turbo-lane (continuous-green-T) screening flags signalized 3-leg T-intersections where one or more main-street through lanes could flow continuously while the minor-street left turn merges in the median — recovering the green time the through movement would otherwise lose to the signal. Candidacy requires measured 3-leg geometry, an arterial-class main street, and a detected median (divided carriageway), all derived from the OpenStreetMap road network. Approach-capacity gain is computed as (turbo lanes / approach lanes) × (1 − g/C) / (g/C), with the main-street through g/C derived from the modeled main-vs-minor critical-flow split; the result is reported within the +7%…+173% envelope documented in the continuous-green-T design literature — the standard national references for this geometry (David Plummer & Associates, Adding Turbo Lanes to T-Intersections, 2010; and the Design Guidelines for the Development of Continuous Green Intersections, 1997). Lane counts and signal timing must be field-verified before design.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \Sigma (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is the NCHRP-716 gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ (home-based-work: a=28507, b=-0.02, c=-0.123) on the travel time t to the signal.

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
Near Rose Avenue	0.74	1.8	10	19.6%
Douglas Boulevard & I 20	0.77	1.8	9	17.6%
Rose Avenue & GA 5	0.86	2.1	6	11.8%
Near I 20	0.90	2.2	4	7.8%
Near GA 5	0.91	2.2	4	7.8%
Signal 20.8mi W of CBD (33.7544, -84.7492)	0.98	2.4	3	5.9%
GA 5 & I 20	0.99	2.4	3	5.9%
Near I 20	1.00	2.4	3	5.9%
Signal 21.3mi W of CBD (33.7259, -84.7572)	1.05	2.5	2	3.9%
GA 5 & I 20	1.07	2.6	2	3.9%
Signal 21.3mi W of CBD (33.7256, -84.7576)	1.08	2.6	2	3.9%
Signal 20.1mi W of CBD (33.7301, -84.7373)	1.10	2.6	2	3.9%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 86% applied via a binary logit $P(auto) = 1 / (1 + e^{-(ASC - \lambda \cdot \Delta GC)})$ calibrated to the metro's measured auto-mode share (ACS B08301) and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 14% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 23.1% of study signals lie on a modeled route; highest loaded-link v/c: 0.56. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Minor arterial	1.52	73	0.56
Freeway / motorway	0.02	42	0.16

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach HCM capacity table. Analysis follows the Highway Capacity Manual 6th Edition signalized-intersection method; v/c, control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

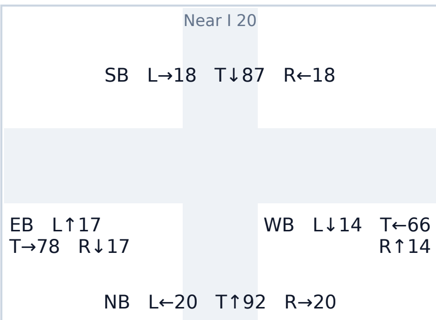
Background turning-movement volumes in the diagrams are distributed from each approach total using an estimated 15/70/15 (Left/Through/Right) split. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

A.1 Near I 20

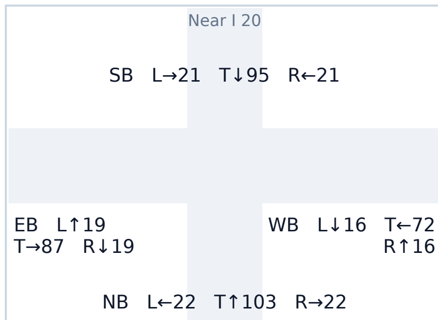
Signal ID	ATL-1917048955
Location	Douglas / West · 1.00 mi from site
Added PM peak trips	3
Existing / No-Build (PM)	LOS B · 16.5 s/veh · v/c 0.28
Build (PM)	LOS B · 16.6 s/veh · v/c 0.29
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

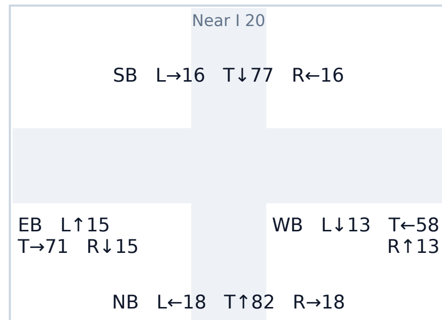
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	146	1	147	0.18 → 0.18	15.4 → 15.4	B → B	91
SB	137	0	137	0.17 → 0.17	15.2 → 15.2	B → B	84
EB	124	2	125	0.15 → 0.15	15.1 → 15.1	B → B	76
WB	104	0	104	0.13 → 0.13	14.8 → 14.8	B → B	63

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Right	1	33%
NB Left	1	33%
EB Through	1	33%

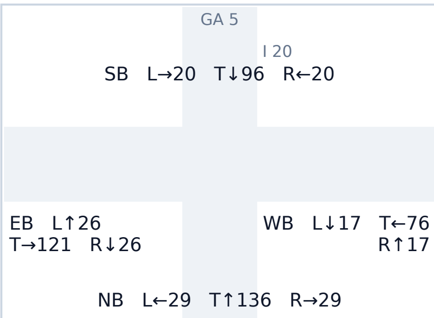
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.2 GA 5 & I 20

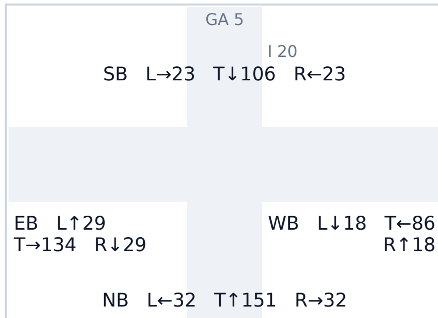
Signal ID	ATL-4562448962
Location	Douglas / West · 1.07 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS B · 17.7 s/veh · v/c 0.38
Build (PM)	LOS B · 17.8 s/veh · v/c 0.38
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

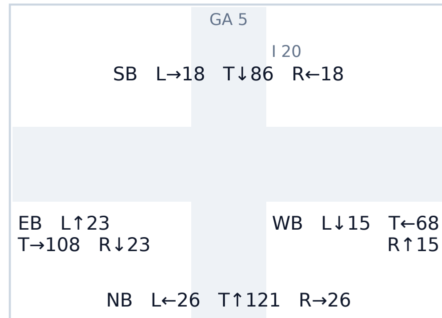
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	215	1	215	0.26 → 0.27	16.3 → 16.3	B → B	139
SB	151	1	152	0.19 → 0.19	15.4 → 15.4	B → B	94
EB	192	0	192	0.24 → 0.24	15.9 → 15.9	B → B	122
WB	122	0	122	0.15 → 0.15	15.0 → 15.0	B → B	74

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	1	50%
NB Through	1	50%

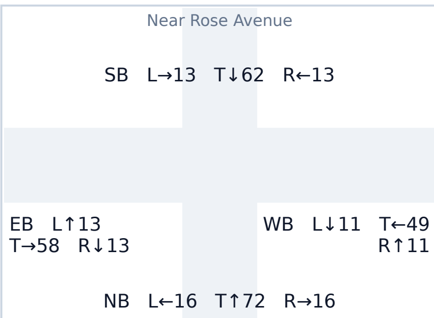
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.3 Near Rose Avenue

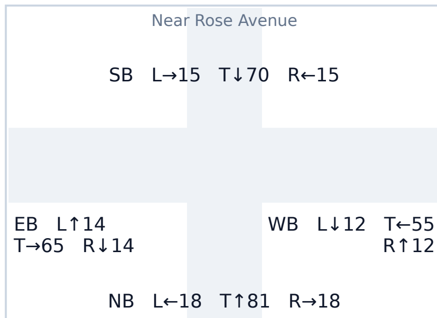
Signal ID	ATL-2174755193
Location	Douglas / West · 0.74 mi from site
Added PM peak trips	10
Existing / No-Build (PM)	LOS B · 15.6 s/veh · v/c 0.21
Build (PM)	LOS B · 15.7 s/veh · v/c 0.22
Δ control delay	+0.1 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

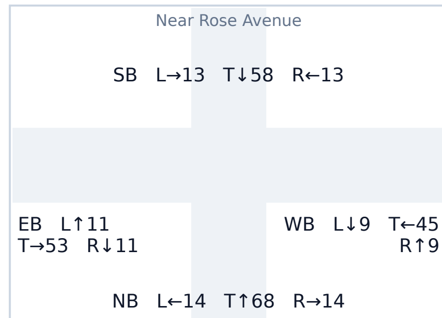
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	112	5	117	0.14 → 0.14	14.9 → 14.9	B → B	71
SB	95	5	100	0.12 → 0.12	14.7 → 14.7	B → B	60
EB	93	0	93	0.11 → 0.11	14.6 → 14.6	B → B	56
WB	79	0	79	0.10 → 0.10	14.5 → 14.5	B → B	47

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	5	50%
NB Through	5	50%

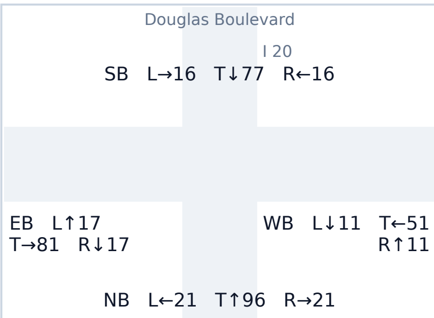
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.4 Douglas Boulevard & I 20

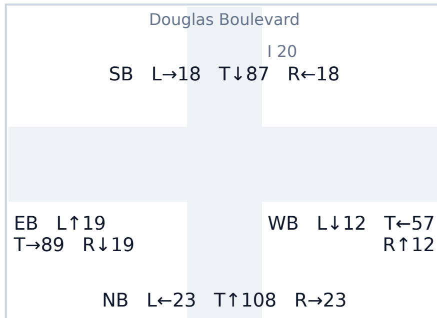
Signal ID	ATL-2884657324
Location	Douglas / West · 0.77 mi from site
Added PM peak trips	9
Existing / No-Build (PM)	LOS B · 16.3 s/veh · v/c 0.27
Build (PM)	LOS B · 16.3 s/veh · v/c 0.27
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

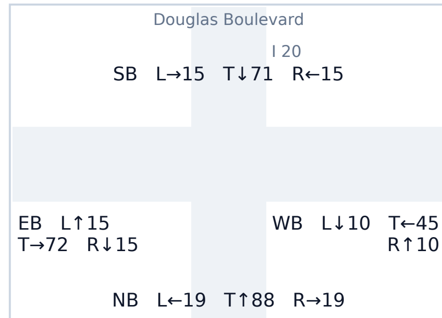
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	151	4	154	0.19 → 0.19	15.4 → 15.4	B → B	96
SB	119	5	123	0.15 → 0.15	15.0 → 15.0	B → B	75
EB	127	0	127	0.16 → 0.16	15.1 → 15.1	B → B	78
WB	80	0	81	0.10 → 0.10	14.5 → 14.5	B → B	48

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	4	44%
NB Through	4	44%
SB Left	1	11%

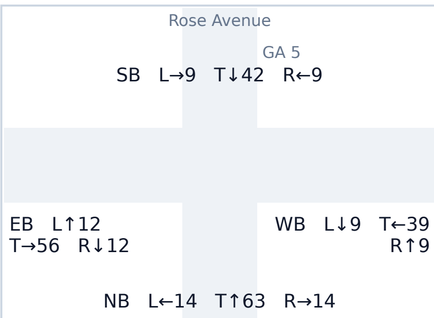
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.5 Rose Avenue & GA 5

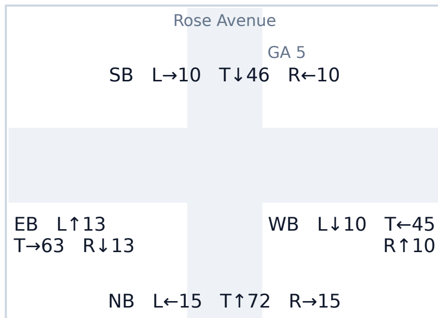
Signal ID	ATL-2174755175
Location	Douglas / West · 0.86 mi from site
Added PM peak trips	6
Existing / No-Build (PM)	LOS B · 15.3 s/veh · v/c 0.18
Build (PM)	LOS B · 15.3 s/veh · v/c 0.18
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

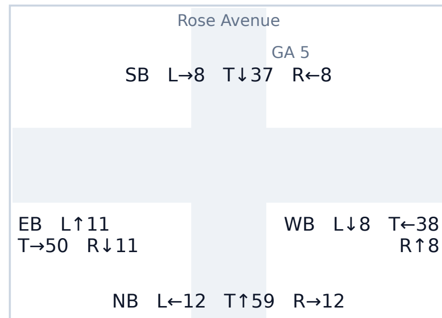
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	99	3	102	0.12 → 0.13	14.7 → 14.7	B → B	61
SB	66	0	66	0.08 → 0.08	14.3 → 14.3	B → B	39
EB	89	0	89	0.11 → 0.11	14.6 → 14.6	B → B	53
WB	62	3	65	0.08 → 0.08	14.3 → 14.3	B → B	38

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Left	3	50%
NB Right	3	50%

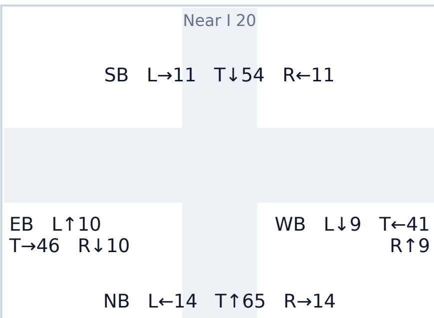
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Near I 20

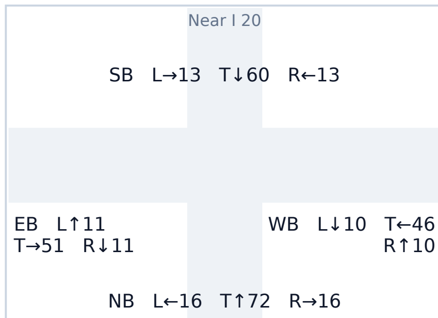
Signal ID	ATL-2884657333
Location	Douglas / West · 0.90 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS B · 15.3 s/veh · v/c 0.18
Build (PM)	LOS B · 15.3 s/veh · v/c 0.18
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

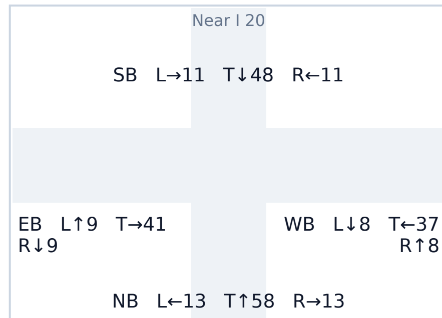
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	102	1	104	0.13 → 0.13	14.8 → 14.8	B → B	63
SB	84	3	86	0.10 → 0.11	14.5 → 14.6	B → B	51
EB	73	0	73	0.09 → 0.09	14.4 → 14.4	B → B	43
WB	66	0	66	0.08 → 0.08	14.3 → 14.3	B → B	39

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	2	50%
NB Through	1	25%
SB Left	1	25%

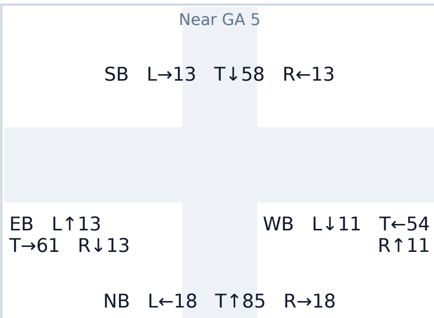
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.7 Near GA 5

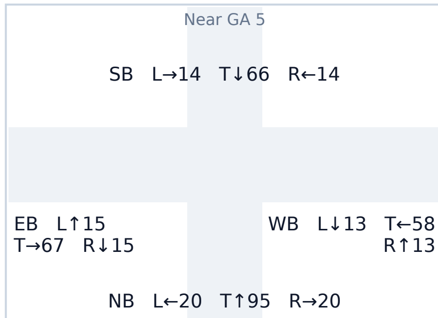
Signal ID	ATL-2884657326
Location	Douglas / West · 0.91 mi from site
Added PM peak trips	4
Existing / No-Build (PM)	LOS B · 15.8 s/veh · v/c 0.23
Build (PM)	LOS B · 15.8 s/veh · v/c 0.23
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

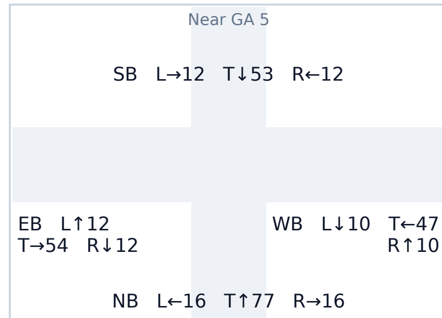
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	133	2	135	0.16 → 0.17	15.1 → 15.2	B → B	83
SB	92	2	94	0.11 → 0.12	14.6 → 14.7	B → B	57
EB	97	0	97	0.12 → 0.12	14.7 → 14.7	B → B	58
WB	84	0	84	0.10 → 0.10	14.5 → 14.5	B → B	50

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	2	50%
NB Through	2	50%

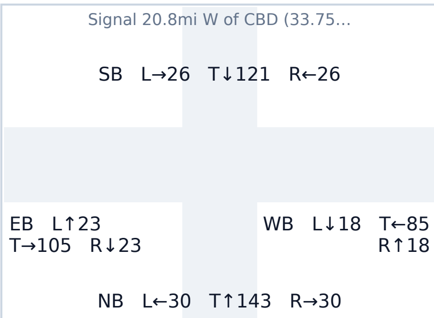
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.8 Signal 20.8mi W of CBD (33.7544, -84.7492)

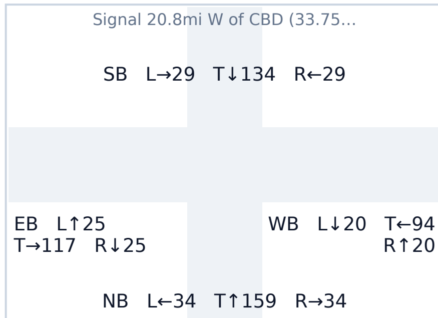
Signal ID	ATL-68429629
Location	Douglas / West · 0.98 mi from site
Added PM peak trips	3
Existing / No-Build (PM)	LOS B · 18.1 s/veh · v/c 0.40
Build (PM)	LOS B · 18.1 s/veh · v/c 0.40
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

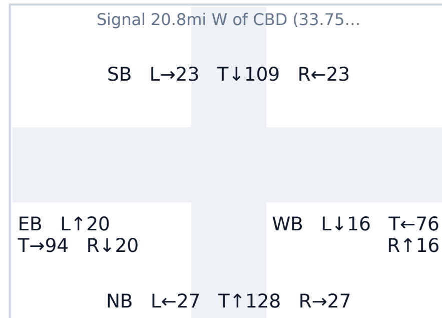
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	225	2	227	0.28 → 0.28	16.4 → 16.4	B → B	147
SB	191	1	192	0.24 → 0.24	15.9 → 15.9	B → B	122
EB	167	0	167	0.21 → 0.21	15.6 → 15.6	B → B	105
WB	134	0	134	0.17 → 0.17	15.2 → 15.2	B → B	82

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
NB Through	2	67%
SB Through	1	33%

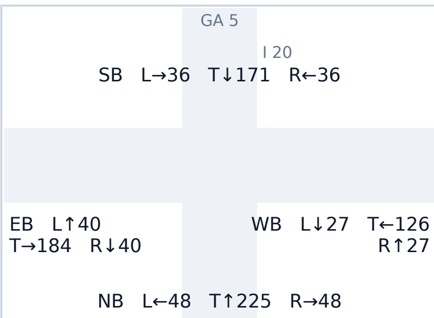
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.9 GA 5 & I 20

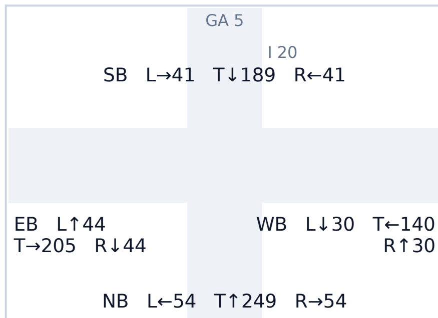
Signal ID	ATL-2884657313
Location	Douglas / West · 0.99 mi from site
Added PM peak trips	3
Existing / No-Build (PM)	LOS C · 22.5 s/veh · v/c 0.62
Build (PM)	LOS C · 22.5 s/veh · v/c 0.62
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

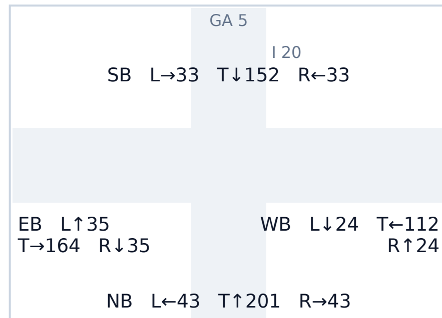
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	356	1	357	0.44 → 0.44	18.7 → 18.7	B → B	253
SB	269	2	271	0.33 → 0.33	17.1 → 17.1	B → B	181
EB	293	0	293	0.36 → 0.36	17.5 → 17.5	B → B	198
WB	200	0	200	0.25 → 0.25	16.0 → 16.0	B → B	127

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	2	67%
NB Through	1	33%

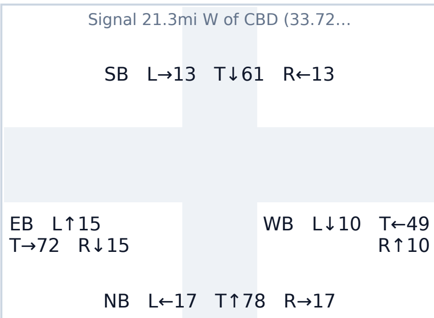
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.10 Signal 21.3mi W of CBD (33.7259, -84.7572)

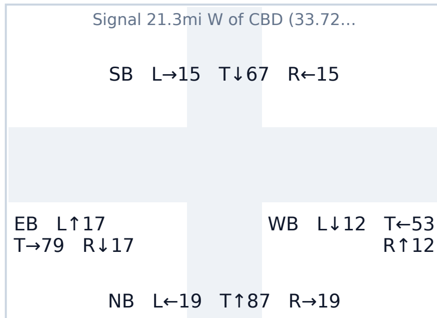
Signal ID	ATL-2884657224
Location	Douglas / West · 1.05 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS B · 15.8 s/veh · v/c 0.23
Build (PM)	LOS B · 15.8 s/veh · v/c 0.23
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

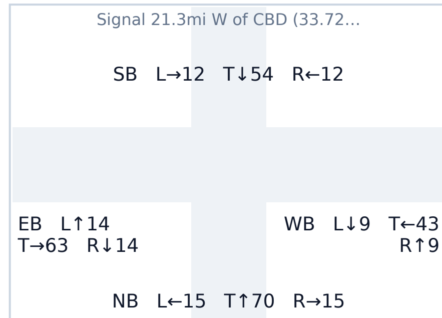
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	124	1	125	0.15 → 0.15	15.0 → 15.0	B → B	76
SB	96	1	97	0.12 → 0.12	14.7 → 14.7	B → B	58
EB	113	0	113	0.14 → 0.14	14.9 → 14.9	B → B	69
WB	77	0	77	0.09 → 0.09	14.5 → 14.5	B → B	46

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	1	50%
NB Through	1	50%

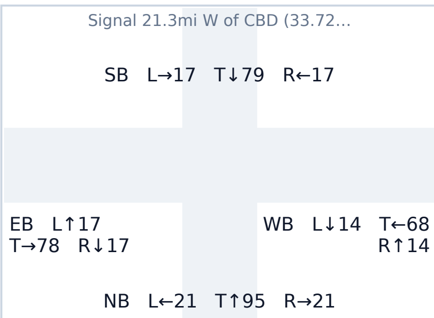
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.11 Signal 21.3mi W of CBD (33.7256, -84.7576)

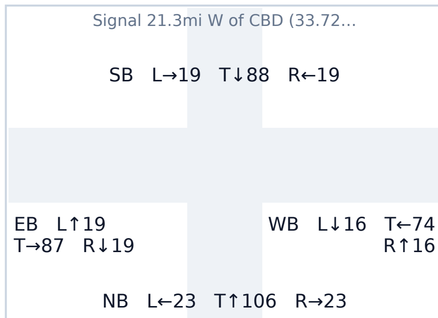
Signal ID	ATL-2884657217
Location	Douglas / West · 1.08 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS B · 16.5 s/veh · v/c 0.28
Build (PM)	LOS B · 16.5 s/veh · v/c 0.28
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

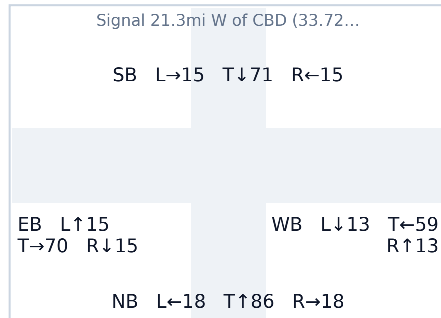
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	151	1	152	0.19 → 0.19	15.4 → 15.4	B → B	94
SB	125	1	126	0.15 → 0.16	15.0 → 15.0	B → B	77
EB	125	0	125	0.15 → 0.15	15.0 → 15.0	B → B	76
WB	106	0	106	0.13 → 0.13	14.8 → 14.8	B → B	64

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	1	50%
NB Through	1	50%

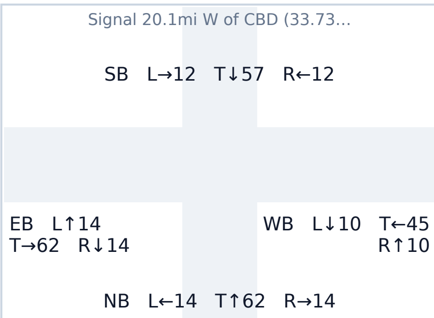
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.12 Signal 20.1mi W of CBD (33.7301, -84.7373)

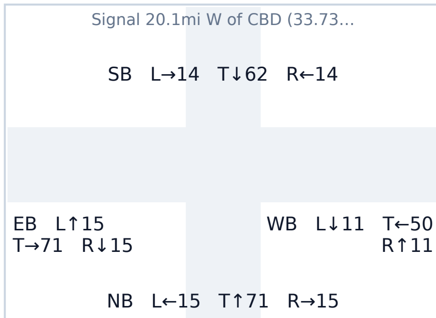
Signal ID	ATL-3128353281
Location	Douglas / West · 1.10 mi from site
Added PM peak trips	2
Existing / No-Build (PM)	LOS B · 15.5 s/veh · v/c 0.20
Build (PM)	LOS B · 15.5 s/veh · v/c 0.20
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

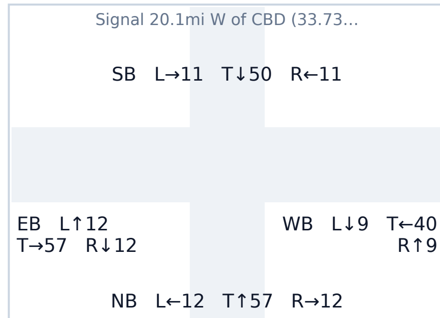
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	100	1	101	0.12 → 0.12	14.7 → 14.7	B → B	60
SB	90	0	90	0.11 → 0.11	14.6 → 14.6	B → B	54
EB	100	1	101	0.12 → 0.12	14.7 → 14.7	B → B	60
WB	72	0	72	0.09 → 0.09	14.4 → 14.4	B → B	42

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Right	1	50%
NB Left	1	50%

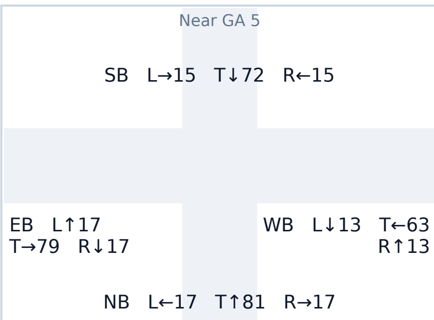
Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.13 Near GA 5

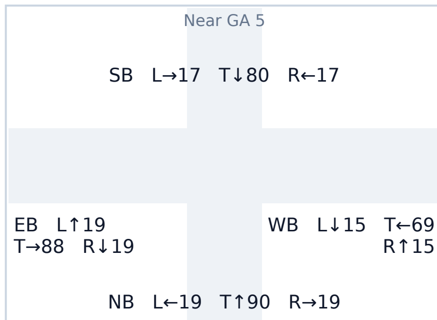
Signal ID	ATL-2884657210
Location	Douglas / West · 1.22 mi from site
Added PM peak trips	1
Existing / No-Build (PM)	LOS B · 16.2 s/veh · v/c 0.26
Build (PM)	LOS B · 16.2 s/veh · v/c 0.26
Δ control delay	+0.0 s/veh
Mitigation	No mitigation required — projected delay change is below the City's 5-second TIS threshold.

Turning-Movement Diagrams (Build condition)

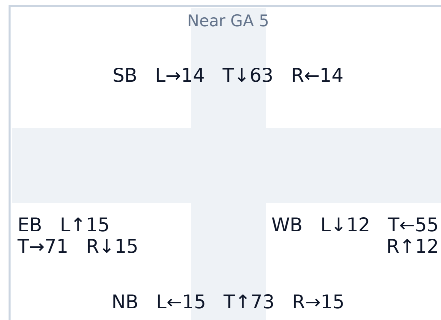
AM Peak Hour — Build



PM Peak Hour — Build



Saturday Midday — Build



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Per-Approach Capacity (PM Peak)

Appr.	Exist vph	+Trips	Build vph	v/c Ex→Bld	Delay Ex→Bld	LOS Ex→Bld	Q95 ft
NB	128	0	128	0.16 → 0.16	15.1 → 15.1	B → B	78
SB	113	1	114	0.14 → 0.14	14.9 → 14.9	B → B	69
EB	126	0	126	0.16 → 0.16	15.0 → 15.0	B → B	77
WB	99	0	99	0.12 → 0.12	14.7 → 14.7	B → B	60

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	1	100%

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.